

Лабораторная Работа №12. Настройка NAT

Администрирование локальных сетей

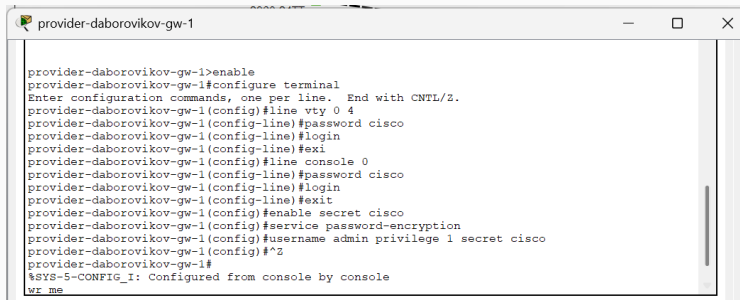
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Приобретение практических навыков по настройке доступа локальной сети к внешней сети посредством NAT.

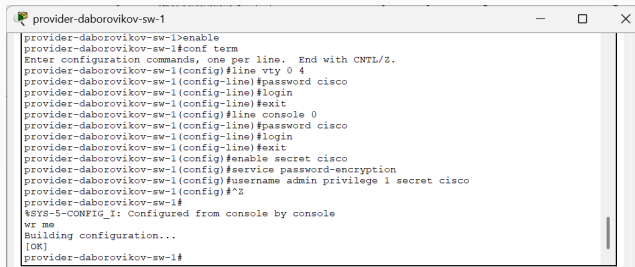
Первоначальная настройка маршрутизатора



```
provider-daborovikov-gw-1>enable
provider-daborovikov-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
provider-daborovikov-gw-1(config)#line vty 0 4
provider-daborovikov-gw-1(config-line)#password cisco
provider-daborovikov-gw-1(config-line)#login
provider-daborovikov-gw-1(config-line)#exit
provider-daborovikov-gw-1(config)#line console 0
provider-daborovikov-gw-1(config-line)#password cisco
provider-daborovikov-gw-1(config-line)#login
provider-daborovikov-gw-1(config-line)#exit
provider-daborovikov-gw-1(config)#enable secret cisco
provider-daborovikov-gw-1(config)#service password-encryption
provider-daborovikov-gw-1(config)#username admin privilege 1 secret cisco
provider-daborovikov-gw-1(config)#^Z
provider-daborovikov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr me
```

Figure 1: Первоначальная настройка маршрутизатора provider-gw-1(присвоение имени, настройка доступа по паролю и др.)

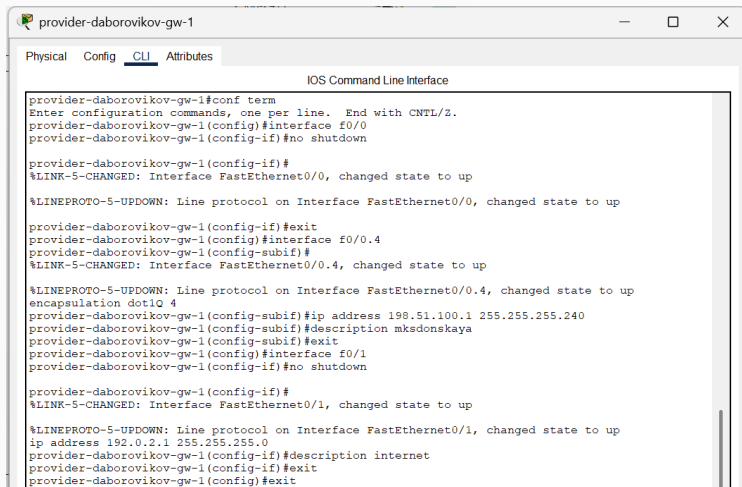
Первоначальная настройка коммутатора



```
provider-daborovikov-sw-1
provider-daborovikov-sw-1>enable
provider-daborovikov-sw-1#conf term
Enter configuration commands, one per line.  End with CNTL/Z.
provider-daborovikov-sw-1(config)#line vty 0 4
provider-daborovikov-sw-1(config-line)#password cisco
provider-daborovikov-sw-1(config-line)#login
provider-daborovikov-sw-1(config-line)#exit
provider-daborovikov-sw-1(config)#line console 0
provider-daborovikov-sw-1(config-line)#password cisco
provider-daborovikov-sw-1(config-line)#login
provider-daborovikov-sw-1(config-line)#exit
provider-daborovikov-sw-1(config)#enable secret cisco
provider-daborovikov-sw-1(config)#service password-encryption
provider-daborovikov-sw-1(config)#username admin privilege 1 secret cisco
provider-daborovikov-sw-1(config)#^Z
provider-daborovikov-sw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr me
Building configuration...
[OK]
provider-daborovikov-sw-1#
```

Figure 2: Первоначальная настройка коммутатора provider-sw-1(присвоение имени, настройка доступа по паролю и др.)

Настройка интерфейсов маршрутизатора



The screenshot shows a window titled 'provider-daborovikov-gw-1' with tabs for 'Physical', 'Config', 'CLI' (selected), and 'Attributes'. The main area displays the 'IOS Command Line Interface' with the following commands and output:

```
provider-daborovikov-gw-1#conf term
Enter configuration commands, one per line. End with CNTL/Z.
provider-daborovikov-gw-1(config)#interface f0/0
provider-daborovikov-gw-1(config-if)#no shutdown

provider-daborovikov-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

provider-daborovikov-gw-1(config-if)#exit
provider-daborovikov-gw-1(config)#interface f0/0.4
provider-daborovikov-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.4, changed state to up

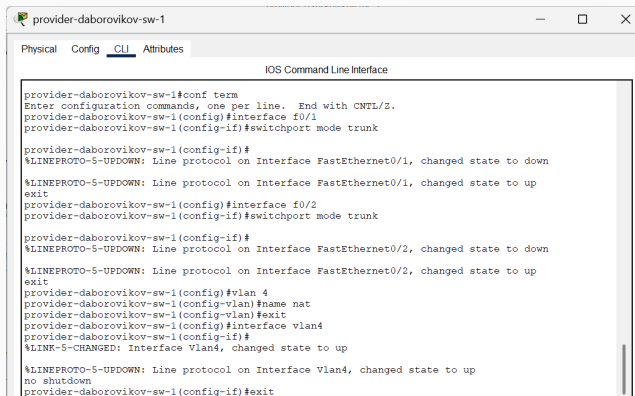
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.4, changed state to up
encapsulation dot1Q 4
provider-daborovikov-gw-1(config-subif)#ip address 198.51.100.1 255.255.255.240
provider-daborovikov-gw-1(config-subif)#description mkstdonskaya
provider-daborovikov-gw-1(config-subif)#exit
provider-daborovikov-gw-1(config)#interface f0/1
provider-daborovikov-gw-1(config-if)#no shutdown

provider-daborovikov-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
ip address 192.0.2.1 255.255.255.0
provider-daborovikov-gw-1(config-if)#description internet
provider-daborovikov-gw-1(config-if)#exit
provider-daborovikov-gw-1(config)#exit
```

Figure 3: Настройка интерфейсов маршрутизатора

Настройка интерфейсов коммутатора



```
provider-daborovikov-sw-1
Physical Config CLI Attributes
IOS Command Line Interface

provider-daborovikov-sw-1#conf term
Enter configuration commands, one per line. End with CNTL/Z.
provider-daborovikov-sw-1(config)#interface f0/1
provider-daborovikov-sw-1(config-if)#switchport mode trunk

provider-daborovikov-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
exit
provider-daborovikov-sw-1(config)#interface f0/2
provider-daborovikov-sw-1(config-if)#switchport mode trunk

provider-daborovikov-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
exit
provider-daborovikov-sw-1(config)#vlan 4
provider-daborovikov-sw-1(config-vlan)#name nat
provider-daborovikov-sw-1(config-vlan)#exit
provider-daborovikov-sw-1(config)#interface vlan4
provider-daborovikov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan4, changed state to up
no shutdown
provider-daborovikov-sw-1(config-if)#exit
```

Figure 4: Настройка интерфейсов коммутатора

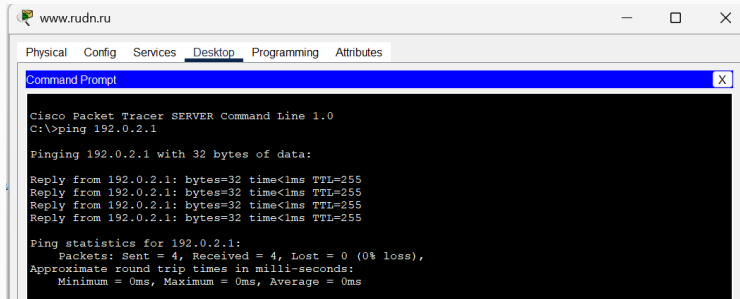
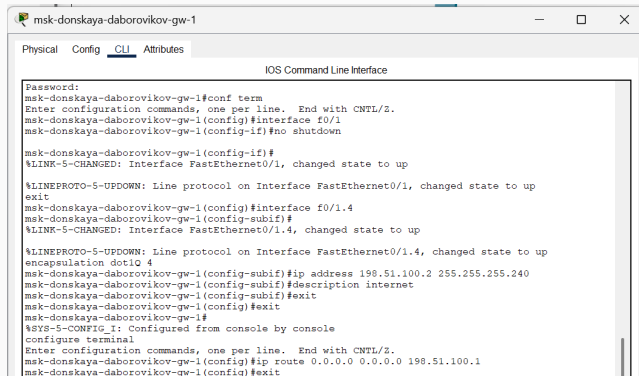


Figure 5: Проверка командой ping с сервера www.rudn.ru на роутер провайдера.

Настройка интерфейсов маршрутизатора



```
msk-donskaya-daborovikov-gw-1
Physical Config CLI Attributes
IOS Command Line Interface

Password:
msk-donskaya-daborovikov-gw-1#conf term
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-gw-1(config)#interface f0/1
msk-donskaya-daborovikov-gw-1(config-if)#no shutdown

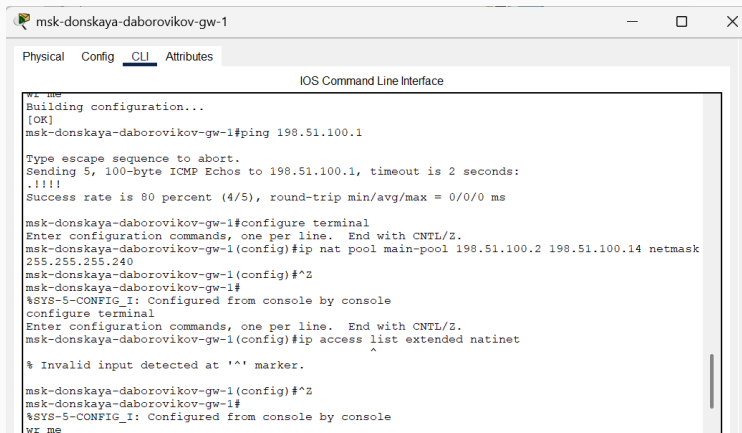
msk-donskaya-daborovikov-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
exit
msk-donskaya-daborovikov-gw-1(config)#interface f0/1.4
msk-donskaya-daborovikov-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/1.4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.4, changed state to up
encapsulation dot1Q 4
msk-donskaya-daborovikov-gw-1(config-subif)#ip address 198.51.100.2 255.255.255.240
msk-donskaya-daborovikov-gw-1(config-subif)#description internet
msk-donskaya-daborovikov-gw-1(config-subif)#exit
msk-donskaya-daborovikov-gw-1(config)#exit
msk-donskaya-daborovikov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-gw-1(config)#ip route 0.0.0.0 0.0.0.0 198.51.100.1
msk-donskaya-daborovikov-gw-1(config)#exit
```

Figure 6: Настройка интерфейсов маршрутизатора msk-donskaya-gw-1 для доступа к сети провайдера.

Настройка списка доступа для NAT



The screenshot shows a terminal window titled 'msk-donskaya-daborovikov-gw-1' with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the 'IOS Command Line Interface'. The user enters 'wr me' to save the configuration. The system prompts 'Building configuration...' and '[OK]'. The user then enters 'ping 198.51.100.1'. The system responds with 'Type escape sequence to abort.', 'Sending 5, 100-byte ICMP Echos to 198.51.100.1, timeout is 2 seconds:', '..!!!!', and 'Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms'. The user enters 'configure terminal'. The system prompts 'Enter configuration commands, one per line. End with CNTL/Z.' The user enters 'ip nat pool main-pool 198.51.100.2 198.51.100.14 netmask 255.255.255.240'. The user enters '^Z'. The system prompts 'msk-donskaya-daborovikov-gw-1(config)#^Z' and 'msk-donskaya-daborovikov-gw-1#'. The system responds with '%SYS-5-CONFIG_I: Configured from console by console' and 'configure terminal'. The user enters 'ip access list extended natinet'. The user enters '^'. The system responds with '% Invalid input detected at '^' marker.' The user enters '^Z'. The system prompts 'msk-donskaya-daborovikov-gw-1(config)#^Z' and 'msk-donskaya-daborovikov-gw-1#'. The system responds with '%SYS-5-CONFIG_I: Configured from console by console' and 'wr me'.

```
msk-donskaya-daborovikov-gw-1#wr me
Building configuration...
[OK]
msk-donskaya-daborovikov-gw-1#ping 198.51.100.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 198.51.100.1, timeout is 2 seconds:
..!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

msk-donskaya-daborovikov-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-gw-1(config)#ip nat pool main-pool 198.51.100.2 198.51.100.14 netmask
255.255.255.240
msk-donskaya-daborovikov-gw-1(config)#^Z
msk-donskaya-daborovikov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-gw-1(config)#ip access list extended natinet
^
% Invalid input detected at '^' marker.

msk-donskaya-daborovikov-gw-1(config)#^Z
msk-donskaya-daborovikov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr me
```

Figure 7: Проверка. Настройка пула адресов для NAT. Настройка списка доступа для NAT

Настройка списка доступа для NAT

```
msk-donskaya-daborovikov-gw-1(config)#ip access-list extended nat-inet
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#remark dk
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#permit tcp 10.128.3.0 0.0.0.255 host 192.0.2.11
eq 80
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#permit tcp 10.128.3.0 0.0.0.255 host 192.0.2.12
eq 80
```

Figure 8: Сеть дисплейных классов (имеют доступ только к сайтам, необходимым для учёбы (www.yandex.ru (192.0.2.11), stud.rudn.university (192.0.2.12))).

Настройка списка доступа для NAT

```
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#remark departments
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#permit tcp 10.128.4.0 0.0.0.255 host 192.0.2.13 eq 80
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#
```

Figure 9: Сеть кафедр (работает только с образовательными сайтами (esystem.pfur.ru (192.0.2.13)))

Настройка списка доступа для NAT

```
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#remark adm  
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#permit tcp 10.128.5.0 0.0.0.255 host 192.0.2.14 eq 80  
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#
```

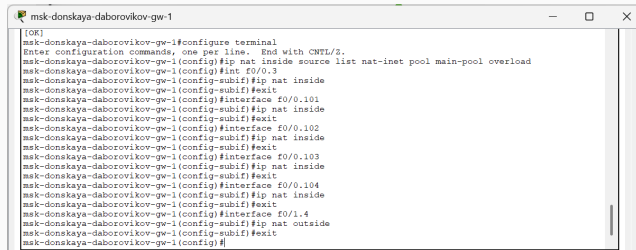
Figure 10: Сеть администрации (имеет возможность работать только с сайтом университета (www.rudn.ru (192.0.2.14)))

Настройка списка доступа для NAT

```
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#remark admin  
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#permit ip host 10.128.6.200 any  
msk-donskaya-daborovikov-gw-1(config-ext-nacl)#
```

Figure 11: Доступ для компьютера администратора (в сети для других пользователей компьютер администратора имеет полный доступ в Интернет. Другие не имеют доступа.)

Настройка NAT



```
msk-donskaya-daborovikov-gw-1
[OK]
msk-donskaya-daborovikov-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-gw-1(config)#ip nat inside source list nat-inet pool main-pool overload
msk-donskaya-daborovikov-gw-1(config)#int f0/0.3
msk-donskaya-daborovikov-gw-1(config-subif)#ip nat inside
msk-donskaya-daborovikov-gw-1(config-subif)#exit
msk-donskaya-daborovikov-gw-1(config)#interface f0/0.101
msk-donskaya-daborovikov-gw-1(config-subif)#ip nat inside
msk-donskaya-daborovikov-gw-1(config-subif)#exit
msk-donskaya-daborovikov-gw-1(config)#interface f0/0.102
msk-donskaya-daborovikov-gw-1(config-subif)#ip nat inside
msk-donskaya-daborovikov-gw-1(config-subif)#exit
msk-donskaya-daborovikov-gw-1(config)#interface f0/0.103
msk-donskaya-daborovikov-gw-1(config-subif)#ip nat inside
msk-donskaya-daborovikov-gw-1(config-subif)#exit
msk-donskaya-daborovikov-gw-1(config)#interface f0/0.104
msk-donskaya-daborovikov-gw-1(config-subif)#ip nat inside
msk-donskaya-daborovikov-gw-1(config-subif)#exit
msk-donskaya-daborovikov-gw-1(config)#interface f0/1.4
msk-donskaya-daborovikov-gw-1(config-subif)#ip nat outside
msk-donskaya-daborovikov-gw-1(config-subif)#exit
msk-donskaya-daborovikov-gw-1(config)#
```

Figure 12: Настройка NAT (Port Address Translation и интерфейсов для NAT).

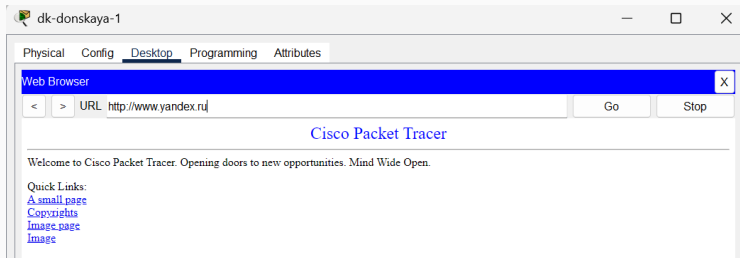
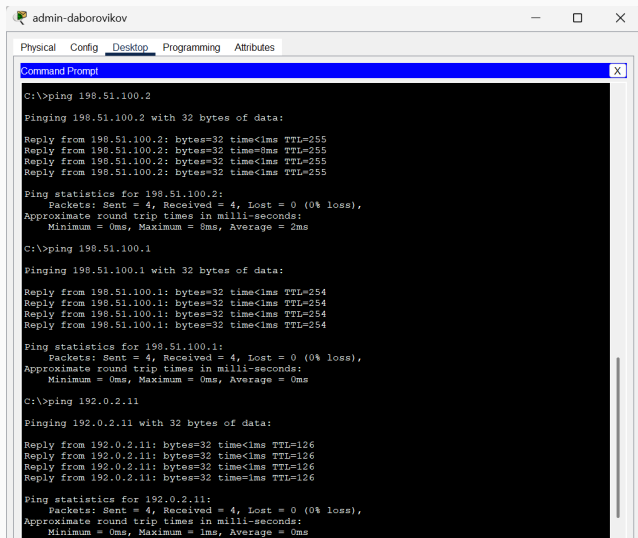


Figure 13: Проверка.



```
admin-daborovikov
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 198.51.100.2

Pinging 198.51.100.2 with 32 bytes of data:

Reply from 198.51.100.2: bytes=32 time<1ms TTL=255
Reply from 198.51.100.2: bytes=32 time=8ms TTL=255
Reply from 198.51.100.2: bytes=32 time<1ms TTL=255
Reply from 198.51.100.2: bytes=32 time<1ms TTL=255

Ping statistics for 198.51.100.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 8ms, Average = 2ms

C:\>ping 198.51.100.1

Pinging 198.51.100.1 with 32 bytes of data:

Reply from 198.51.100.1: bytes=32 time<1ms TTL=254
Reply from 198.51.100.1: bytes=32 time<1ms TTL=254
Reply from 198.51.100.1: bytes=32 time<1ms TTL=254
Reply from 198.51.100.1: bytes=32 time<1ms TTL=254

Ping statistics for 198.51.100.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.0.2.11

Pinging 192.0.2.11 with 32 bytes of data:

Reply from 192.0.2.11: bytes=32 time<1ms TTL=126
Reply from 192.0.2.11: bytes=32 time<1ms TTL=126
Reply from 192.0.2.11: bytes=32 time<1ms TTL=126
Reply from 192.0.2.11: bytes=32 time=1ms TTL=126

Ping statistics for 192.0.2.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Figure 14: Проверка.

```
msk-donskaya-daborovikov-gw-1#conf term
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-gw-1(config)#ip nat inside source static tcp 10.128.0.2 80 198.51.100.2
80
msk-donskaya-daborovikov-gw-1(config)#
```

Figure 15: Настройка доступа из Интернета (WWW-сервер).

```
msh-donskaya-daborovikov-gw-1(config)#ip nat inside source static tcp 10.128.0.3 20 198.51.100.3 20  
msh-donskaya-daborovikov-gw-1(config)#ip nat inside source static tcp 10.128.0.3 21 198.51.100.3 21  
msh-donskaya-daborovikov-gw-1(config)#
```

Figure 16: Настройка доступа из Интернета (файловый сервер).

Доступ из внешней сети в локальную

```
msh-donskaya-daborovikov-gw-1(config)#ip nat inside source static tcp 10.128.0.4 25 198.51.100.4 25  
msh-donskaya-daborovikov-gw-1(config)#ip nat inside source static tcp 10.128.0.4 110 198.51.100.4 110  
msh-donskaya-daborovikov-gw-1(config)#
```

Figure 17: Настройка доступа из Интернета (почтовый сервер).

Доступ из внешней сети в локальную

```
msk-donskaya-daborovikov-gw-1(config)#ip nat inside source static tcp 10.128.6.200 3389 198.51.100.10 3389
msk-donskaya-daborovikov-gw-1(config)#^Z
msk-donskaya-daborovikov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr me
Building configuration...
[OK]
msk-donskaya-daborovikov-gw-1#
```

Figure 18: Настройка доступа из Интернета (доступ по RDP).

Доступ из внешней сети в локальную

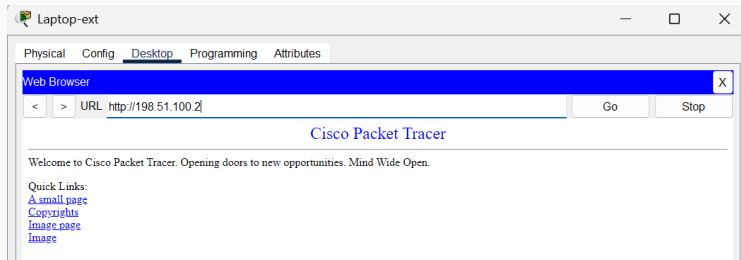


Figure 19: Проверка.

В ходе выполнения лабораторной работы я приобрел практические навыки по настройке доступа локальной сети к внешней сети посредством NAT.